

Lecture 5

FAIR and Open Science

Principles, Reproducibility, and Open Research

Computing for Data-Driven Biology · 5BI00A
Master's Programme in Bioinformatics · Umeå University

Session Aims & ILO Links

- 1 State and explain the four FAIR principles with a concrete bioinformatics example for each
- 2 Describe the open science landscape — open access, open data, open source, open peer review
- 3 Explain why Git is reproducibility infrastructure, not just version control
- 4 Describe FAIR4RS and explain how it extends FAIR to software and code
- 5 Connect open science practices to the UN Sustainable Development Goals
- 6 Identify the criteria for the essay assessment and describe what VG-level work looks like

ILO 3

ILO 3

ILO 3, 12

ILO 3

ILO 3

ILO 13

What does FAIR stand for?

Word cloud — type the four words (one at a time)

Also: Have you ever tried to reanalyse or reproduce published bioinformatics data?

Yes — successfully

Yes — but hit problems

No — tried to find data

No — never attempted

And: word cloud — in one word, why does reproducibility matter in science?

Part 1

The reproducibility problem — and why it matters in bioinformatics

The Reproducibility Problem: Evidence

This is not a fringe concern — it is a structural problem with measurable consequences.

70%

of researchers have failed to reproduce another scientist's experiment

Baker, Nature (2016), n=1,576

50%

have failed to reproduce their own experiments

Baker, Nature (2016)

<50%

of high-profile RNA-seq papers were computationally reproducible in a 2022 systematic study

Wijesooriya et al. (2022)

Consequences in bioinformatics:

Wasted public funding · Clinical decisions based on unreproducible results · Literature pollution through citations of flawed work · Career consequences for the researchers involved

The Causes — and Their Solutions

Irreproducibility in bioinformatics is largely preventable. Each cause has a direct solution.

✗ Undescribed software versions

✓ Pin exact versions in methods; use containers

✗ Unavailable raw data

✓ Deposit in ENA/SRA/GEO with a persistent accession

✗ Undocumented parameters

✓ Specify all non-default parameters; commit scripts to Git

✗ Code never version-controlled

✓ Git from day one; tag commits that generate manuscript results

✗ Statistical reporting without pipeline

✓ Provide analysis code; workflow manager (Snakemake/Nextflow)

✗ Format and dependency issues

✓ Containers (Docker/Singularity) package code + environment together

Part 2

The FAIR principles — what they mean in practice

The FAIR Principles: An Overview

Wilkinson et al. (2016) *Scientific Data* 3:160018 · DOI: 10.1038/sdata.2016.18

FAIR is primarily about enabling machines to find, access, integrate, and reuse data — not just humans.

F

Findable

Persistent identifiers · Rich metadata · Indexed in searchable resources

A

Accessible

Standardised retrieval protocol · Authentication where needed · Metadata persists even if data is removed

I

Interoperable

Standard formats · Ontology-based metadata · Cross-references to related resources

R

Reusable

Rich metadata · Data licence · Provenance · Community standards (MINSEQE, MixS)

FAIR ≠ open. Controlled-access data can be fully FAIR. The opposite of FAIR is not 'restricted' — it is 'unusable by machines.'

Findable and Accessible: In Practice

Findable

F

Persistent identifiers (DOIs, accession numbers, ORCIDs) · Rich metadata indexed in searchable databases · The same dataset accessible from ENA, GEO, or ArrayExpress via a stable accession

- ✓ Raw reads in ENA with accession PRJEB12345, samples annotated with EFO ontology terms
- ✗ Data in supplementary Excel file — no identifier, not indexed, lost if publisher changes hosting

Accessible

A

Retrievable by identifier via standardised protocol (HTTPS, FTP) · Authentication documented where needed · Metadata persists as a tombstone record even if data must be withdrawn

- ✓ GTEx: raw data via documented dbGaP application; processed summaries fully open via stable URL
- ✗ 'Data available upon reasonable request' — corresponding author has since left academia

Interoperable and Reusable: In Practice

I Interoperable

Standard formats (FASTQ, BAM, VCF, GFF) · Ontology-based metadata (GO, EFO, NCBI Taxonomy, UBERON) · Cross-references to related datasets · APIs with documented endpoints

- ✓ Sample annotated as UBERON:0002107 (liver) + EFO:0001272 (treated) — machine-readable, comparable across studies
- ✗ Sample annotated as 'liver (adult, treated with drug X)' — unstructured, cannot be automatically integrated

R Reusable

Rich metadata sufficient to assign samples without author contact · Data licence (CC-BY, CC0) · Full provenance chain · Software version pinning · Community metadata standard declared

- ✓ Sample table with taxonomy ID, tissue ontology, treatment, time point; CC-BY licence; software versions specified; Snakemake workflow in Zenodo with DOI
- ✗ Samples labelled 'control' and 'treated'; no licence; 'standard parameters used'; code available upon request

Think of a bioinformatics paper you have read — or one from your undergraduate degree.

1. Was the raw data deposited somewhere you could find with a persistent identifier?

2. Could you have told which samples were which from the metadata alone?

3. Was the analysis code available — and if so, were software versions specified?

4. Could you reproduce the main result from what was published?

Share back: what was the most common failure mode in your examples?



Break — 10 minutes

Back at: _____

After the break: open science · Git as reproducibility · FAIR4RS · SDGs · essay assessment

Part 3

Open science · Git as reproducibility · FAIR4RS · Essay

The Open Science Landscape

Open Access

Gold OA: publisher makes final version immediately open (often APC required)

Green OA: author self-archives in repository (PubMed Central, bioRxiv)

Diamond OA: no cost to author or reader

Preprints: bioRxiv/medRxiv — rapid sharing before peer review

Vetenskapsrådet and ERC mandate OA for funded research.

Open Data

Data in public repositories with persistent identifiers and rich metadata

General repositories: Zenodo, Figshare

Domain-specific: ENA, GEO, PRIDE, ArrayExpress

FAIR principles apply directly here

Data management plans now required by most funders

Open Source Software

Source code available under an open licence (MIT, Apache 2.0, GPL)

Tools you already use: BLAST+, samtools, bcftools, bedtools — all open source

Licences matter: without one, code cannot legally be reused

Resource: choosealicense.com

Open Peer Review

Reviewer identities and/or review content made public

Increases transparency and accountability

Adopted by eLife, PeerJ, F1000Research, and others

Preprint review platforms: Review Commons, PREREVIEW

Git as Reproducibility Infrastructure

You have been using Git for version control. It is also — when used well — the backbone of reproducible science.



Every commit is a reproducibility checkpoint

The complete history of every analysis file is preserved. When a reviewer asks 'exactly what did you do?', you can show the exact code at the time of submission.



Tags connect commits to publications

Create a tag (e.g. v1.0-paper-submission) at the commit used to generate manuscript results. Reference this tag in the methods section. Readers can retrieve that exact version.



Archive repositories beyond GitHub

GitHub repositories can be deleted. For publications, archive via Zenodo (integrates directly with GitHub — each tagged release gets a DOI automatically) or Software Heritage.



Containers extend reproducibility to the environment

Code alone is insufficient — the software environment matters too. Docker or Singularity containers package code + all dependencies together. This is FAIR4RS in practice.

FAIR for Research Software (FAIR4RS)

Chue Hong et al. (2022) · DOI: 10.15497/RDA00068

The same four principles — applied to software and code.

F

Findable

✗ Script emailed as attachment on request, no identifier

✓ GitHub repo with Zenodo DOI, registered in bio.tools

A

Accessible

✗ Compiled binary only, no source code

✓ Source code on GitHub with stable URL; Zenodo archive for permanence

I

Interoperable

✗ Reads and writes proprietary formats, no documented API

✓ Standard I/O formats (FASTA, VCF), documented CLI and API

R

Reusable

✗ No licence, no README, no version history, ad hoc variable names

✓ MIT licence, README with install instructions, pinned dependencies, version tags

Open Science and the UN Sustainable Development Goals

Open science is not merely good academic practice — it is an instrument of global equity and environmental sustainability.

SDG 3

Good Health & Well-Being

Open access to genomic, clinical, and pharmacogenomic data accelerates medical research and enables researchers in low- and middle-income countries to participate in global science.

SDG 14

Life Below Water

eDNA metabarcoding data deposited in ENA using SILVA and BOLD reference databases enables global marine biodiversity monitoring. Open data is the raw material of conservation policy.

SDG 15

Life on Land

GBIF, BOLD, SILVA, and UNITE are open data infrastructures directly used by CBD monitoring frameworks. Biodiversity assessment at scale is only possible because these databases are open and FAIR.

SDG 17

Partnerships for the Goals

Open science is itself a mechanism of international partnership — open data, open access, and open software enable researchers everywhere to build on shared knowledge, regardless of institutional resources.

Essay Assessment

Introducing the FAIR compliance essay

The FAIR Essay: What It Asks

Full details, marking criteria, deadline, and submission instructions are in Canvas.

The task

Critically evaluate the FAIR compliance of a provided published bioinformatics study and its associated data. Your evaluation should cover all four FAIR principles, the sufficiency of metadata for reuse, and the reproducibility of the computational analysis. Make specific, evidenced recommendations.

G**Pass**

Correctly identifies and describes FAIR issues · Applies principles accurately · Notes data deposition and code availability · Observations are relevant and accurate

VG**Distinction**

All of the above, plus: Evaluates consequences of each FAIR issue · Makes specific, concrete recommendations · Distinguishes FAIR spirit from letter · Structured, evidenced, analytical — not merely descriptive

What to Look For: The Assessment Checklist

F

- ☐ Accession number in the paper? Is it persistent (SRA/ENA/GEO/DOI) or just a URL?
- ☐ Can you find the dataset via a database search engine without knowing the accession?

A

- ☐ Can you actually download the data via HTTPS or FTP?
- ☐ If access-controlled, is the application process documented?

I

- ☐ Are data in standard formats (FASTQ, BAM, VCF)?
- ☐ Are samples annotated with ontology terms or free text?
- ☐ Is the reference genome version specified with a versioned identifier?

What a Strong Response Looks Like

G level

"The raw data was deposited in the NCBI SRA with accession SRP123456 (F). The data can be downloaded (A). The authors used standard FASTQ format (I). The methods section describes the analysis (R)."

Describes facts correctly but does not evaluate quality, consequences, or what could have been improved.

VG level

"While raw reads were deposited in SRA (SRP123456), Findability is compromised: no EFO annotations were deposited for the six conditions, making semantic discovery impossible without prior knowledge of the accession. Regarding Reusability: HISAT2 v2.1.0 and DESeq2 v1.20.0 are specified, but exact parameters are omitted and no code is available — 'standard parameters were used' is insufficient for reproduction. This is remedied straightforwardly by depositing a MINSEQE-compliant sample table and committing analysis scripts to a versioned repository at submission."

What You Should Know After This Session



State all four FAIR principles and give a concrete bioinformatics example for each



Explain the difference between accessible and open — controlled access can be FAIR



Explain why ontology terms make metadata more interoperable than free-text annotations



Describe three components of open science and name one key resource for each



Explain how Git functions as reproducibility infrastructure beyond version control



Describe FAIR4RS and give an example of the difference between FAIR and non-FAIR software



Apply the FAIR framework to a bioinformatics study, identifying issues and their consequences



Describe what distinguishes a VG-level essay response from a G-level one